

Practice Test: Quadratic Factorization

1. Factor: $x^2 + 6x + 8$

Answer: $(x + 2)(x + 4)$

2. Factor: $x^2 - 16$

Answer: $(x - 4)(x + 4)$

3. Solve by factoring: $x^2 - 8x + 15 = 0$

Answer: $x = 3$ or $x = 5$

4. Factor: $2x^2 + 7x + 3$

Answer: $(2x + 1)(x + 3)$

5. Factor: $x^2 - x - 12$

Answer: $(x - 4)(x + 3)$

6. Solve: $x^2 + 2x - 8 = 0$

Answer: $x = 2$ or $x = -4$

7. Factor: $3x^2 - 12$

Answer: $3(x - 2)(x + 2)$

8. Factor: $x^2 + 10x + 25$

Answer: $(x + 5)^2$

9. Solve: $4x^2 - 9 = 0$

Answer: $x = \frac{3}{2}$ or $x = -\frac{3}{2}$

10. Factor: $x^2 - 5x - 14$

Answer: $(x - 7)(x + 2)$